

Achieve asymptotically three-quarters optimal impartial selection in single-nomination graphs

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Improved Bounds for Single-Nomination Impartial Selection*.

1 The open problem

This paper addresses an open problem posed by Javier Cembrano, Felix Fischer, Max Klimm in *Improved Bounds for Single-Nomination Impartial Selection* [1] (Economics and Computation (EC), 2023), Section 5 Upper Bounds, page 14 (end of section introduction) and also Section 1.1 Our Contribution, page 3 (end). The website catalogue entry is problem 15 (W4383558145_p0) of the [Open Problems in Operations Research](#) collection.

Determine whether there exists a family of impartial selection mechanisms $(f_n)_{n \geq 1}$, with f_n defined on G_n , such that

$$\liminf_{n \rightarrow \infty} \inf_{G \in G_n: \Delta(G) > 0} \frac{\mathbb{E}_{v \sim f_n(G)}[\delta^-(v, G)]}{\Delta(G)} \geq \frac{3}{4}.$$

Equivalently, decide whether for every $\varepsilon > 0$ there exists n_0 and impartial mechanisms f_n (for all $n \geq n_0$) satisfying

$$\inf_{G \in G_n: \Delta(G) > 0} \frac{\mathbb{E}_{v \sim f_n(G)}[\delta^-(v, G)]}{\Delta(G)} \geq \frac{3}{4} - \varepsilon.$$

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

Status. Partial progress, not a complete solution. The normalization and impartiality issues can be repaired for several new degree regimes, but we still do not obtain a single mechanism that works simultaneously on every graph.

Throughout, let G^{-S} denote the graph obtained by deleting the outgoing arcs of the vertices in S .

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_050949_c88ed499, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

2 Exact completion of impartial submechanisms

We call q an *impartial submechanism* if $q_i(G) \geq 0$, if q_i is independent of i 's outgoing arc, and if

$$\sum_i q_i(G) \leq 1.$$

There is a general exact completion: choose a default vertex d uniformly at random, run q on G^{-d} , and assign the unused mass to d . Equivalently,

$$M_i(G) = \frac{1}{n} \sum_d q_i(G^{-d}) + \frac{1}{n} \left(1 - \sum_j q_j(G^{-i}) \right). \quad (1)$$

Then M is nonnegative and normalized. It is also impartial: by the impartiality of q , every term $q_i(G^{-d})$ with $d \neq i$ is invariant under changes to i 's arc, and the remaining terms are evaluated on G^{-i} .

Suppose q has welfare at least c on every partial graph of maximum indegree D . Fix a maximum-degree vertex v^* in G . For the $n - D$ defaults that do not nominate v^* , deleting the default's edge preserves the maximum indegree D . Hence

$$\mathbb{E}_M[d_{\text{sel}}] \geq c \left(1 - \frac{D}{n} \right). \quad (2)$$

This lemma supplies exact normalization without any graph-dependent renormalization.

3 A Lipschitz-score construction

Let $\phi : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be nondecreasing and satisfy

$$0 \leq \phi(k) - \phi(k-1) \leq L.$$

Define

$$q_i^\phi(G) = \frac{\phi(d_i(G))}{L + \sum_v \phi(d_v(G^{-i}))}. \quad (3)$$

This is impartial. Put

$$S = \sum_v \phi(d_v(G)).$$

Deleting one outgoing edge decreases S by at most L , so every denominator in (3) is at least S . Consequently,

$$\sum_i q_i^\phi(G) \leq 1. \quad (4)$$

Thus the completion (1) turns q^ϕ into an exact normalized impartial mechanism.

3.1 Every fixed $D \geq 8$

Fix $D \geq 8$ and put

$$c = \frac{3D}{4}, \quad t = \lceil c \rceil - 1, \quad \phi_D(k) = (k - t)_+. \quad (5)$$

Here $L = 1$. Write

$$S = \sum_v \phi_D(d_v), \quad A = \sum_v d_v \phi_D(d_v).$$

Since deletion only decreases S , (3) gives

$$\sum_i d_i q_i^{\phi_D}(G) \geq \frac{A}{1 + S}. \quad (6)$$

Moreover,

$$A - c(1 + S) = \sum_v (d_v - c) \phi_D(d_v) - c. \quad (7)$$

Every term in the sum is nonnegative, because positive ϕ_D -weight requires $d_v \geq \lceil c \rceil$. A vertex of degree D alone contributes

$$(D - c)(D - t) = \frac{D}{4}(D - \lceil 3D/4 \rceil + 1) \geq \frac{3D}{4} \quad (8)$$

for every $D \geq 8$. Therefore

$$\sum_i d_i q_i^{\phi_D}(G) \geq \frac{3D}{4}. \quad (9)$$

The exact completion (1) of q^{ϕ_D} satisfies

$$\boxed{\frac{\mathbb{E}[d_{\text{sel}}]}{D} \geq \frac{3}{4} \left(1 - \frac{D}{n}\right) \quad (\Delta(G) = D \geq 8).} \quad (10)$$

This settles every promised fixed degree $D \geq 8$, but the mechanism depends on the promised value D .

4 The promised degree $D = 4$

We use

$$\phi(k) = (k - 2)_+, \quad L = 1. \quad (11)$$

Let

$$U = \{v : d_v \geq 3\}, \quad h = |U|,$$

and let m_j count the vertices of degree j ; thus $m_4 \geq 1$. Put

$$S = 2m_4 + m_3, \quad A = 8m_4 + 3m_3. \quad (12)$$

Call a default *exceptional* if it lies in U or nominates a vertex of U . There are at most

$$h + \sum_{v \in U} d_v \leq 5h \quad (13)$$

such defaults.

For a nonexceptional default, the ϕ -profile is unchanged. Let

$$I = \{i \in U : i \rightarrow U\}, \quad K = \sum_{i \in I} \phi(d_i), \quad C = \sum_{i \in I} d_i \phi(d_i).$$

Candidates in I have denominator S , while the other candidates in U have denominator $S + 1$. Therefore the submechanism's total mass and welfare are

$$R = \frac{K}{S} + \frac{S - K}{S + 1}, \quad W_q = \frac{C}{S} + \frac{A - C}{S + 1}. \quad (14)$$

Its deficit is

$$\rho = 1 - R = \frac{S - K}{S(S + 1)}. \quad (15)$$

If the default has mean indegree 1, then

$$W_q + \rho - 3 = \frac{2m_4 - 2 + (C - K)/S}{S + 1} \geq 0. \quad (16)$$

If $h = o(n)$, the nonexceptional defaults have mean indegree $1 - o(1)$, because only $O(h)$ vertices were excluded and all indegrees are at most 4. Equations (14)–(16) therefore give welfare $3 - o(1)$.

If h is not $o(n)$, then after deleting any one edge the total positive ϕ -weight tends to infinity. The submechanism then places mass $1 - o(1)$ on vertices of degree at least 3, again giving welfare $3 - o(1)$.

For example, splitting at $h = \sqrt{n}$ yields

$$\boxed{\inf_{\Delta(G)=4} \frac{\mathbb{E}[d_{\text{sel}}]}{4} \geq \frac{3}{4} - O(n^{-1/2}).} \quad (17)$$

5 A further universally defined restricted mechanism

There is also a simple mechanism for graphs whose unique maximum is separated from the runner-up by at least two. Define

$$g_i(G) = \mathbf{1} \left\{ d_i(G) \geq 2 + \max_{v \neq i} d_v(G^{-i}) \right\}. \quad (18)$$

This is impartial, and at most one coordinate can equal 1: if both $g_i = g_j = 1$, then

$$d_i \geq d_j + 1, \quad d_j \geq d_i + 1,$$

a contradiction.

If x is the unique vertex of maximum degree D and every other degree is at most $D - 2$, then $g_x = 1$. After exact default completion, all defaults but the D predecessors of x still produce x . Hence

$$\boxed{\frac{\mathbb{E}[d_{\text{sel}}]}{D} \geq 1 - \frac{D}{n}} \quad (19)$$

on this subclass. Thus the unresolved fixed-degree cases are specifically the tied or one-gap configurations, not all sparse-maximum configurations.

6 The omitted threshold-mechanism impartiality argument

For the original degree-two construction,

$$r_{di} = \frac{\mathbf{1}\{d_i(G^{-d}) \geq 2\}}{1 + |\{v : d_v(G^{-\{d,i\}}) \geq 2\}|},$$

and

$$f_i(G) = \frac{1}{n} \left(1 - \sum_{j \neq i} r_{ij}(G) + \sum_{d \neq i} r_{di}(G) \right). \quad (20)$$

Every r_{ij} deletes i 's outgoing edge. In r_{di} , the denominator also deletes i 's edge, while the numerator depends only on i 's indegree, which cannot depend on i 's outgoing edge because loops are forbidden. Hence every term in (20) is invariant under changes to i 's nomination. This fully repairs the omitted impartiality proof.

The previously established bound therefore remains

$$\inf_{\Delta(G)=2} \frac{\mathbb{E}[d_{\text{sel}}]}{2} \geq \frac{3}{4} - \frac{1}{n}. \quad (21)$$

7 Why the scalar construction does not repair $D = 3$

The preceding Lipschitz construction has a sharp degree-three obstruction.

Take a realizable degree sequence with one vertex of degree three, two vertices of degree two, four vertices of degree zero, and all remaining vertices of degree one. Arrange the three vertices of degree two or three so that they nominate degree-one vertices. Such loopless realizations exist for every sufficiently large n .

Normalize $L = 1$ and suppose first that $\phi(0) = \phi(1) = 0$. Write

$$x = \phi(2), \quad y = \phi(3).$$

The Lipschitz condition implies

$$0 \leq x \leq 1, \quad x \leq y \leq x + 1. \quad (22)$$

For all but $O(1)$ defaults, the completed mechanism's welfare tends to

$$W(x, y) = \frac{1 + 3y + 4x}{1 + y + 2x}. \quad (23)$$

This expression is increasing in y , and after setting $y = x + 1$,

$$W(x, x + 1) = \frac{4 + 7x}{2 + 3x},$$

which is increasing in x . Thus

$$W(x, y) \leq W(1, 2) = \frac{11}{5} < \frac{9}{4}. \quad (24)$$

Nonnegligible weight on degree-one vertices only pulls the ratio toward 1. Hence no member, mixture, or pointwise oracle choice from this scalar Lipschitz family repairs degree 3.

This is an obstruction to the proposed construction family, not an impossibility theorem for arbitrary impartial mechanisms.

8 The gluing problem remains unrepaired

Let A, B be normalized impartial mechanisms. A graph-level adaptive mixture

$$F_i(G) = (1 - \lambda(G))A_i(G) + \lambda(G)B_i(G)$$

is impartial across an i -edge change $G \leftrightarrow G'$ only if

$$(\lambda(G) - \lambda(G'))(B_i(G) - A_i(G)) = 0. \quad (25)$$

Thus switching according to $\Delta(G)$ generally fails impartiality.

Candidatewise switching restores impartiality:

$$q_i(G) = (1 - \lambda(\Delta(G^{-i})))A_i(G) + \lambda(\Delta(G^{-i}))B_i(G). \quad (26)$$

But suppose v is the unique vertex of maximum degree D . Exactly its predecessors see maximum degree $D - 1$ after deleting their own edges. Therefore

$$\sum_i q_i(G) = 1 + (\lambda(D - 1) - \lambda(D)) \sum_{i \rightarrow v} (B_i(G) - A_i(G)), \quad (27)$$

which need not equal 1.

Dividing by $\sum_i q_i(G)$ does not repair this: q_i is constant along an i -edge fiber, but the denominator generally is not, so the normalized i -probability ceases to be impartial.

If $|\lambda(k) - \lambda(k - 1)| \leq \eta$, then comparison with the fixed mixture at $\lambda(D)$ gives

$$\left| \sum_i q_i - 1 \right| \leq 2\eta. \quad (28)$$

Thus slowly varying transitions over a growing degree window can be completed with only $O(\eta)$ loss. This does not, however, bridge adjacent fixed degrees such as 2 and 3, where the required transition has order-one slope.

Final disposition

The verifier's fundamental rejection is only partially repaired.

The strongest new rigorous conclusions are:

promised class	guarantee
$\Delta = 1$	1
$\Delta = 2$	$\frac{3}{4} - \frac{1}{n}$
$\Delta = 4$	$\frac{3}{4} - O(n^{-1/2})$
$\Delta = D \geq 8$ fixed	$\frac{3}{4} \left(1 - \frac{D}{n} \right)$
unique maximum with gap ≥ 2	$1 - \frac{D}{n}$

The remaining blocking issues are:

1. no construction here covers every graph of fixed maximum degree $D \in \{3, 5, 6, 7\}$;
2. the promised- D mechanisms cannot be switched according to the observed maximum degree without violating impartiality or normalization;
3. consequently, no single normalized impartial family with a simultaneous $3/4 - o(1)$ guarantee over all $G \in G_n$ has been constructed.

The finite upper bound

$$\alpha_n^* \leq \frac{3}{4} - \frac{1}{4n} + \frac{1}{n(n-2)(n-4)}$$

remains correct and tends to $3/4$, so it does not settle the asymptotic question negatively. Thus the canonical result remains unresolved in this attempt.

References

- [1] Javier Cembrano, Felix Fischer, Max Klimm. *Improved Bounds for Single-Nomination Impartial Selection*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597693>